Deep Learning for Crowd Modelling: Behavioural Analysis and Pedestrian Trajectory Prediction — Literature Review

Nov 10, 2026 · @Aakriti

Status and reading guide

Phase 1 deliverable. No code has been written and no model has been trained; every performance number below is quoted from a published paper or benchmark and attributed to it.

Every substantive statement carries one of three labels:

- **[Verified]** — taken from the cited primary source (paper, official repository or publisher record), with a DOI or link in Section 10.
- **[Interpretation]** — our reading of what verified findings mean for this project.
- **[Recommendation]** — a proposed decision for the next phase, open to supervisor approval.

Reference numbers in square brackets, e.g. [3], follow IEEE style and point to Section 10.

Update, 11 Nov 2026: the project builds on a professor-provided baseline. How this review feeds that baseline, and what to learn before implementation, is in Research strategy (baseline-first).

1. Introduction

This project asks how deep learning can predict where pedestrians in a crowd will walk next, and when modelling their interactions actually improves those predictions.

Motivation. Railway stations concentrate large, time-varying crowds on platforms, stairs and concourses. Reliable short-horizon prediction of pedestrian motion supports crowd monitoring, congestion warning, facility design and evacuation planning. Indian stations have recorded dangerous overcrowding, including a stampede-like situation at New Delhi station on 15 February 2025 [36].

The prediction task [Verified]. Given the observed positions of every pedestrian in a scene over a past window, predict each pedestrian's future positions. The common benchmark protocol observes 8 frames (3.2 s) and predicts 12 frames (4.8 s) at 0.4 s per frame [6], [7].

What makes it hard [Verified]. Surveys identify five core challenges: motion uncertainty, interaction modelling, scene understanding, data limitations and interpretability [30]. Several plausible futures exist for one past (multimodality), and a good prediction must also be physically acceptable, for example free of collisions [14].

Scope. Input is pedestrian trajectories (positions over time). Raw-video detection and tracking is out of scope for Phase 1 and remains an open question for the supervisor.

Research questions (draft, [Recommendation]).

1. RQ1 — How much do learned models improve on non-learned baselines (constant velocity, linear, social force)?
2. RQ2 — Which interaction mechanism (none, grid pooling, attention/graph, generative) gives the best balance of accuracy, physical plausibility and cost?
3. RQ3 — How does the benefit of interaction modelling vary with crowd density and interaction type?
4. RQ4 — How well do models trained on low-density benchmarks transfer to denser, station-like scenes?

2. Background

Crowd science gives deep learning its targets and its baselines: physics-based models describe how pedestrians move, and learned models try to predict it from data.

2.1 Levels of crowd modelling

- **Macroscopic** models treat the crowd as a continuum (density and velocity fields), like a fluid.
- **Microscopic** models simulate each pedestrian as an agent with its own position, velocity and goal. Social force, velocity-obstacle and all deep trajectory predictors in this review are microscopic.
- **[Interpretation]** Trajectory prediction is a microscopic task; crowd-level quantities (density, flow, congestion) are then aggregated from individual predictions.

2.2 Pedestrian dynamics and the fundamental diagram

The core empirical law of pedestrian flow is the **fundamental diagram**: walking speed falls as density rises [4]. With density ρ (persons/m²) and mean speed v (m/s), the specific flow is J = ρ · v (persons/m/s).

```
J(\rho) = \rho \, v(\rho)
```

[Verified] Weidmann's frequently cited diagram summarised 25 earlier investigations and reaches a maximum density of 5.4 persons/m²; some later studies measured higher densities [4]. Seyfried et al. found single-file and planar fundamental diagrams agree closely, suggesting lateral interference has little effect on the speed–density relation [4].

[Interpretation] For railway platforms this matters directly: a prediction model trained at low density learns near free-flow speeds and may over-predict speed when the platform is congested.

2.3 Social force model

[Verified] Helbing and Molnár model each pedestrian as if driven by "social forces" that express internal motivations rather than physical contact [1]. Pedestrian α accelerates towards a desired velocity, is repelled by other pedestrians β and by borders B, and can be attracted by objects; the resulting equations are nonlinearly coupled Langevin equations [1]:

```
\frac{d\mathbf{v}_\alpha}{dt} = \frac{1}{\tau_\alpha}\left(v^0_\alpha
\mathbf{e}_\alpha - \mathbf{v}_\alpha\right) + \sum_{\beta \neq \alpha}
\mathbf{f}_{\alpha\beta} + \sum_{B} \mathbf{f}_{\alpha B} +
\boldsymbol{\xi}_\alpha
```

Here v^0 is the desired speed, e the desired direction, τ a relaxation time and ξ fluctuations. Repulsive terms are gradients of potentials that decrease with distance [1]. Simulations reproduce several observed self-organisation effects [1]. An escape-panic variant added physical contact forces [2], and many later variants exist [40].

2.4 Collision avoidance

[Verified] Velocity-obstacle methods choose a velocity that avoids collision within a time horizon. Optimal Reciprocal Collision Avoidance (ORCA) gives each agent half the responsibility for avoiding each pairwise collision and reduces the choice of velocity to a low-dimensional linear program, with a sufficient condition for collision-free motion [5].

[Verified] A cognitive alternative uses two visual heuristics for desired direction and speed; combined with body collisions it reproduces lane formation and the crowd turbulence seen in crowd disasters at extreme densities [3].

2.5 Congestion and collective movement

[Verified] Collective patterns — lanes in counter-flow, stop-and-go waves, turbulence at extreme density — emerge from local interactions between individuals [3].
[Interpretation] These patterns are what a good interaction model should reproduce; they also define behaviour categories for our analysis.

2.6 Crowd density estimation

[Verified] In video, crowd density is usually estimated by regressing a density map whose integral over any region gives the count of people in it [28]. The Multi-Column CNN (MCNN) uses three CNN columns with different filter sizes to cope with head-size variation due to perspective [29].

[Interpretation] Our input is trajectories, so density can be computed directly from positions (persons per m² in a region) instead of estimated from images. Density-map estimation becomes relevant only if raw video is brought into scope.

2.7 Data-driven crowd simulation and behaviour in public spaces

[Verified] Example-based simulation learns behaviour from recorded trajectories instead of hand-written rules; Lerner et al.'s "Crowds by Example" is the origin of the UCY data [9], [8]. In train stations, Alahi et al. studied 42 million trajectories and forecast pedestrians' destinations using origin–destination priors [27].

[Interpretation] Deep trajectory predictors are a form of data-driven simulation: run autoregressively, they can roll a crowd forward in time. Physics models remain the reference for dense crowds, where data are scarce (Section 7).

3. Literature review

The field moved from hand-crafted forces (1995) to recurrent social pooling (2016), generative models (2018–2022), graph and attention models (2018–2021) and Transformers (2020–2021), with physics-informed hybrids and diffusion models as current directions. Each entry gives venue and identifier; full IEEE references are in Section 10. Numbers are quoted only from the cited source.

3.1 Recurrent models with social pooling

[6] Social LSTM: Human Trajectory Prediction in Crowded Spaces — A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, S. Savarese. CVPR 2016, pp. 961–971. DOI 10.1109/CVPR.2016.110.

- **Objective:** learn human–human interactions from data instead of hand-crafted social forces.
- **Architecture [Verified]:** one LSTM per pedestrian; a social pooling layer lets spatially close LSTMs share hidden states through an $N_o \times N_o$ grid around each person; the next position is a bivariate Gaussian (μx, μy, σx, σy, ρ) trained by negative log-likelihood. Variants: Vanilla LSTM (no pooling) and O-LSTM (pools occupancy only).

- **Data and metrics:** ETH and UCY; average displacement error, final displacement error and average non-linear displacement error.
- **Findings [Verified]:** reported more than 42% improvement over previous forecasting methods; the advantage was largest on the denser UCY scenes.
- **Limitations [Verified]:** later reimplementation under the Social GAN protocol reports average ADE/FDE of 0.72/1.54 m for Social LSTM against 0.70/1.52 m for Vanilla LSTM on ETH/UCY [7]; the Trajectron authors note this contradiction between two papers by the same group [16]. **[Interpretation]** The grid has fixed size and resolution, and per-agent recurrence is slow.
- **Relevance:** the classical baseline for interaction modelling and the starting point of this project's comparison.

[7] Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks — A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, A. Alahi. CVPR 2018, pp. 2255–2264. DOI 10.1109/CVPR.2018.00240.

- **Objective:** predict several socially plausible futures, because human motion is multimodal.
- **Architecture [Verified]:** LSTM encoder–decoder generator with a pooling module (MLP followed by max pooling over all people), trained adversarially against a recurrent discriminator, with a "variety loss" that rewards the best of k samples.
- **Data and metrics:** ETH and UCY (5 sets, 4 scenes, 1536 pedestrians); ADE and FDE, evaluated by sampling several times and keeping the best prediction.
- **Findings [Verified]:** authors report better accuracy, variety, collision avoidance and speed than prior work; reported ETH/UCY average with 20 samples is 0.58/1.18 m.
- **Limitations [Verified]:** best-of-N evaluation makes real-world use unclear, since the lowest-error sample is unknown at run time [16].
- **Relevance:** its data splits and preprocessing became the de-facto ETH/UCY protocol used by later codebases.

3.2 Attention and graph models

[10] Social Attention: Modeling Attention in Human Crowds — A. Vemula, K. Muelling, J. Oh. ICRA 2018, pp. 4601–4607.

- **Objective / architecture [Verified]:** a spatio-temporal graph of RNNs with soft attention that weighs each neighbour by learned importance, not by proximity alone.
- **Data:** two public crowd datasets (ETH, UCY).
- **Findings [Verified]:** more accurate than the state-of-the-art approach of the time; attention weights can be inspected.
- **Relevance:** the simplest attention replacement for grid pooling.

[11] STGAT: Modeling Spatial-Temporal Interactions for Human Trajectory Prediction — Y. Huang, H. Bi, Z. Li, T. Mao, Z. Wang. ICCV 2019, pp. 6272–6281. DOI 10.1109/ICCV.2019.00637.

- **Architecture [Verified]:** sequence-to-sequence model; graph attention captures spatial interactions at each step and an extra LSTM encodes how interactions evolve in time.
- **Data and metrics:** ETH and UCY; ADE/FDE.
- **Limitations [Verified]:** the authors' repository states that the paper's description of ADE is wrong and should read RMSE/L2 distance as in Social GAN — an example of metric inconsistency in the field.
- **Relevance:** a strong, well-documented attention baseline built on Social GAN data.

[12] Social-STGCNN: A Social Spatio-Temporal Graph Convolutional Neural Network for Human Trajectory Prediction — A. Mohamed, K. Qian, M. Elhoseiny, C. Claudel. CVPR 2020, pp. 14424–14432. DOI 10.1109/CVPR42600.2020.01443.

- **Architecture [Verified]:** models pedestrians as a spatio-temporal graph, with a kernel function that embeds social interactions in the adjacency matrix, replacing aggregation (pooling) layers.
- **Findings [Verified]:** 20% better FDE than prior state of the art, 8.5× fewer parameters, up to 48× faster inference, and it exceeded previous ADE using only 20% of the training data.
- **Relevance:** the most compute-efficient interaction model; attractive under limited GPU access.

3.3 Generative multimodal models

[16] The Trajectron (B. Ivanovic, M. Pavone, ICCV 2019) and **[13] Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data** — T. Salzmann, B. Ivanovic, P. Chakravarty, M. Pavone. ECCV 2020, LNCS, pp. 683–700. DOI 10.1007/978-3-030-58523-5_40.

- **Architecture [Verified]:** modular graph-structured recurrent model with a conditional variational autoencoder (CVAE); integrates agent dynamics so outputs are dynamically feasible, and can use semantic maps.
- **Data and metrics:** ETH/UCY and nuScenes; best-of-N ADE/FDE and KDE negative log-likelihood.
- **Findings [Verified]:** outperformed Social GAN and Social BiGAT on all three metrics on ETH/UCY.
- **Relevance:** reference generative model; official code, models and data are public.

[20] It Is Not the Journey but the Destination: Endpoint Conditioned Trajectory Prediction (PECNet) — K. Mangalam et al. ECCV 2020, LNCS 12347, pp. 759–776. DOI 10.1007/978-3-030-58536-5_45.

- **Architecture [Verified]:** first infers a distribution over distant endpoints, then predicts paths conditioned on them; uses a non-local social pooling layer.
- **Findings [Verified]:** about 20.9% improvement on Stanford Drone and 40.8% on ETH/UCY over the prior state of the art.
- **Relevance:** goal/destination conditioning maps naturally onto stations, where exits, stairs and doors are fixed destinations.

[21] Stochastic Trajectory Prediction via Motion Indeterminacy Diffusion (MID) — T. Gu et al. CVPR 2022, pp. 17113–17122. DOI 10.1109/CVPR52688.2022.01660.

- **Architecture [Verified]:** casts prediction as reversing a diffusion process, progressively removing uncertainty; Transformer-based denoiser conditioned on a history and interaction encoding.
- **Limitations [Verified]:** iterative sampling is slow; the default uses 100 diffusion steps, and the authors' 2-step accelerated version gives 0.41/0.71 on ETH against 0.39/0.66 with 100 steps.
- **Relevance:** representative of the current diffusion frontier; likely too costly for this project's core.

3.4 Transformer models

[17] Transformer Networks for Trajectory Forecasting — F. Giuliari, I. Hasan, M. Cristani, F. Galasso. ICPR 2020. arXiv:2003.08111.

- **Architecture [Verified]:** a standard Transformer per person with no explicit interaction terms.
- **Findings [Verified]:** best score on the TrajNet benchmark; its multimodal extension performs on par with more engineered methods on ETH/UCY.
- **Relevance [Interpretation]:** strong evidence that sequence modelling alone is a hard baseline — interaction modules must beat it to justify their cost.

[18] Spatio-Temporal Graph Transformer Networks for Pedestrian Trajectory Prediction (STAR) — C. Yu, X. Ma, J. Ren, H. Zhao, S. Yi. ECCV 2020. DOI 10.1007/978-3-030-58610-2_30.

- **Architecture [Verified]:** attention only — spatial graph Transformers (TGConv) interleaved with temporal Transformers, plus an external graph memory.
- **Findings [Verified]:** outperformed prior models on 4 of 5 ETH/UCY sets and was comparable on the fifth.

[19] AgentFormer: Agent-Aware Transformers for Socio-Temporal Multi-Agent Forecasting — Y. Yuan, X. Weng, Y. Ou, K. Kitani. ICCV 2021, pp. 9813–9823. DOI 10.1109/ICCV48922.2021.00967.

- **Architecture [Verified]:** flattens all agents and time steps into one sequence; agent-aware attention keeps agent identity; latent intents are modelled jointly (CVAE).
- **Limitations [Verified]:** the official repository reports a normalisation bug; after fixing it, performance is worse than the ICCV paper, and corrected numbers are in the updated arXiv version.
- **Relevance:** state-of-the-art design, but a clear warning to re-verify published numbers.

3.5 Physics-informed hybrids

[22] Human Trajectory Prediction via Neural Social Physics (NSP) — J. Yue, D. Manocha, H. Wang. ECCV 2022. arXiv:2207.10435.

- **Architecture [Verified]:** a neural differential equation containing an explicit social-force-style model whose parameters are learned by neural networks.
- **Findings [Verified]:** improved on 15 recent deep methods across 6 datasets by 5.56%–70%, and kept plausible predictions at 2–5× the training density.
- **Relevance:** the best documented route to interpretable, density-robust prediction; fits a mechanical-engineering framing.

[23] SFMGNet: A Physics-based Neural Network to Predict Pedestrian Trajectories — S. Hossain, F. T. Johora, J. P. Müller, S. Hartmann, A. Reinhardt. arXiv:2202.02791, 2022.

- **Architecture [Verified]:** social force model extended by a group force, combined with an MLP; considers obstacles, pedestrians and groups.
- **Limitation [Interpretation]:** preprint; initial results only.

3.6 Benchmarks and critical analyses

[14] Human Trajectory Forecasting in Crowds: A Deep Learning Perspective (TrajNet++) — P. Kothari, S. Kreiss, A. Alahi. IEEE Trans. Intell. Transp. Syst., 2021. DOI 10.1109/TITS.2021.3069362.

- **Contribution [Verified]:** an interaction-centric benchmark with a hierarchy of trajectory categories and collision-based metrics for physical feasibility.
- **Findings [Verified]:** on interacting trajectories of the real-world split, LSTM 0.60/1.30 m with 13.6% collisions vs S-LSTM 0.53/1.14 m with 6.7% (mean of 5 runs).
- **Relevance [Interpretation]:** reconciles the Social-LSTM debate — pooling helps on interaction-heavy cases and on collisions even when average ADE barely moves.

[15] What the Constant Velocity Model Can Teach Us About Pedestrian Motion Prediction — C. Schöller, V. Aravantinos, F. Lay, A. Knoll. IEEE RA-L, vol. 5, no. 2, 2020. DOI 10.1109/LRA.2020.2969925.

- **Findings [Verified]:** a constant velocity model outperformed state-of-the-art neural models; networks implicitly learn environment priors that hurt generalisation.
- **Relevance:** a mandatory baseline.

3.7 Emerging directions (context only)

[24] Can Language Beat Numerical Regression? Language-Based Multimodal Trajectory Prediction (LMTraj) — I. Bae, J. Lee, H.-G. Jeon. CVPR 2024. Recasts prediction as question answering for a language model and reports state-of-the-art results on public pedestrian benchmarks [Verified]. **[Interpretation]** Out of scope: heavy compute and weak link to crowd physics.

4. Architecture comparison

No single family dominates on every axis: recurrent pooling is simple and interpretable, graph models are cheapest, generative and diffusion models capture uncertainty best, and physics-informed hybrids generalise best to higher density. Method descriptions come from Section 3 **[Verified]**; the cost, data and fit columns are **[Interpretation]**.

Family	Examples	How it predicts	Interaction modelling	Cost / data need	Main strength	Main limitation	Fit for this project
Kinematic baselines	Constant velocity, linear [15]	Extrapolates last velocity	None	Negligible / none	Hard to beat on short horizons	Ignores everyone else	Required baseline
Physics-based	Social force [1], ORCA [5]	Integrates forces or solves an LP for velocity	Hand-crafted repulsion and attraction	Low / few calibrated parameters	Interpretable; valid at high density	Fixed functional forms; tuning needed	Required baseline
Recurrent, no interaction	Vanilla LSTM [6]	LSTM over own past; Gaussian or point output	None	Low / moderate	Simple, stable	Blind to neighbours	Core baseline
Recurrent + grid pooling	Social LSTM, O-LSTM [6]	Per-agent LSTM; neighbours' states pooled on a grid	Local grid of hidden states or occupancy	Medium (per-step pooling) / moderate	Direct, inspectable interaction term	Fixed grid; mixed reproduction record	Core candidate
Generative (GAN)	Social GAN [7]	Encoder–decoder generator + discriminator	Global MLP + max pool	Medium–high; training can be unstable	Multiple futures	Best-of-N evaluation	Comparison only

Family	Examples	How it predicts	Interaction modelling	Cost / data need	Main strength	Main limitation	Fit for this project
Generative (CVAE)	Trajectron++ [13], PECNet [20]	Latent variable model, sampled futures; PECNet conditions on endpoints	Graph edges (Trajectron++); non-local pooling (PECNet)	Medium / moderate	Strong accuracy, dynamics-aware, goal-aware	More moving parts	Strong candidate (one model)
Attention / graph attention	Social Attention [10], STGAT [11]	RNN + attention weights over neighbours	Learned importance per neighbour	Medium / moderate	Weights are interpretable	Attention ≠ explanation by itself	Strong candidate
Graph convolution	Social-STGCNN [12]	Spatio-temporal graph conv, kernel-weighted adjacency	Graph adjacency	Low; data-efficient	8.5× fewer parameters, fast	Fixed kernel choice	Strong candidate
Transformer	TF [17], STAR [18], AgentFormer [19]	Self-attention over time (and agents)	None (TF) to joint socio-temporal attention	High / large	State-of-the-art capacity	Data-hungry; reproducibility issues [19]	Optional comparison
Diffusion	MID [21]	Iterative denoising from noise to trajectory	Encoded state embedding	High at inference	Rich uncertainty	Slow sampling	Future work
Physics-informed hybrid	NSP [22], SFMGNet [23]	Learned parameters inside a force model / ODE	Explicit social forces	Medium / moderate	Interpretable; density-robust	Fewer public reproductions	Strong candidate (extension)
Language models	LMTraj [24]	Trajectory as text question answering	Implicit via prompts	Very high	Semantic reasoning	Compute, latency	Out of scope

[Recommendation, revised 11 Nov 2026] The project builds on the professor's baseline, so this table is context for understanding and discussing it, not a replacement plan. Kinematic baselines (constant velocity, linear) should be reported beside it; any further comparison model is proposed only after the baseline has been inspected and only with the professor's approval.

5. Dataset research

ETH/UCY is the unavoidable benchmark but is sparse; Grand Central Station and the Czech station dataset are the closest public matches to railway stations. Facts in the table are **[Verified]** from the cited sources; the last column is **[Recommendation]**.

Dataset	Source	Setting	Size	Density	Access	Role here
ETH (ETH, Hotel)	Pellegrini et al. 2009 [8]	Outdoor, high vantage point	Part of 5 sets, 1536 pedestrians total with UCY [7]	ETH 0.148 ped/m² [35]	Public; Social GAN preprocessed splits widely used	Primary benchmark
UCY (Univ, Zara1, Zara2)	Lerner et al. 2007 [9]	Campus and shopping street	See above	Univ 0.391, Zara1 0.206, Zara2 0.267 ped/m² [35]	Public	Primary benchmark
TrajNet++	Kothari et al. 2021 [14]	ETH/UCY and other real scenes + synthetic	Interaction-categorised scenes	Mixed	Public with baselines	Interaction-type analysis (RQ2, RQ3)
Stanford Drone (SDD)	Robicquet et al. 2016 [25]	University campus, drone view	20,000 targets in over 100 top-view scenes; pedestrians, cyclists, skaters, cars	Mixed	CC BY-NC-SA 3.0	Secondary; pixel coordinates
Grand Central Station	Zhou et al. 2012 [26]; annotations Yi et al. 2015 [43]	New York station concourse, surveillance view	Walking routes of 12,684 pedestrians from a one-hour video [43]	About 80–100 persons per frame on average [35]	Original site offline; links kept in the CIDNN repository	Main station-like test (RQ4)
Czech train terminals	Apeltauer et al. 2024 [39]	Railway stations, Havlíčkův Brod and Pardubice	Over 24 million trajectory samples	To be measured	CC BY 4.0	Candidate second station dataset
Lyon Festival of Lights	Korbmacher & Tordeux 2024 [37]	Dense public event	Mean 55 tracked pedestrians	0.2–2.2 ped/m², mean 0.95	Check availability	Density-band evaluation
Train-station trajectories	Alahi et al. 2014 [27]	Train stations	42 million trajectories	High	Availability not confirmed	Reference only
CrowdTraj	arXiv 2026 [38]	CCTV dense crowds	Peak 114–372 pedestrians per frame	Very high	New benchmark	Out of scope unless video is in scope

[Verified] Why density matters. Most field datasets cover only low-density, free-flow conditions; deep models thrive at low density while physics-based models excel at high density [37]. A 2025 review states that datasets of truly high-density crowds are still missing [35].

[Interpretation] Protocol details to standardise. The common ETH/UCY protocol samples at 0.4 s (2.5 fps), observes 8 steps and predicts 12, and uses leave-one-scene-out evaluation [7]. Grand Central needs a pixel-to-metre homography before metric ADE/FDE is meaningful.

6. Mathematical foundations

Every model in the shortlist is built from the same pieces: a recurrent or attention encoder of each person's past, a neighbourhood operator that mixes in other people, an output distribution, a loss and an optimiser.

6.1 Problem notation

Scene with N pedestrians. Pedestrian i has position $p_i^t = (x_i^t, y_i^t)$ in metres at time step t. Observe t = 1 … T_obs (8 steps), predict t = T_obs + 1 … T_pred (12 steps). Inputs are often relative displacements $\Delta p_i^t = p_i^t - p_i^{t-1}$; **[Verified]** predicting such residuals reduces error margins compared with predicting absolute positions [15].

6.2 Recurrent neural networks

A simple RNN updates a hidden state h_t from the previous state and the current input x_t:

```
h_t = \tanh(W_h h_{t-1} + W_x x_t + b)
```

Repeated multiplication by W_h over many steps makes gradients vanish or explode, which motivated the LSTM [32].

6.3 LSTM: gates, hidden state and cell state

The LSTM [32] adds a cell state c_t, a memory line updated additively, and gates that control it. In the now-standard form (with a forget gate):

$$
\begin{aligned}
f_t &= \sigma(W_f [h_{t-1}, x_t] + b_f) && \text{forget gate} \\
i_t &= \sigma(W_i [h_{t-1}, x_t] + b_i) && \text{input gate} \\
g_t &= \tanh(W_g [h_{t-1}, x_t] + b_g) && \text{candidate memory} \\
o_t &= \sigma(W_o [h_{t-1}, x_t] + b_o) && \text{output gate} \\
c_t &= f_t \odot c_{t-1} + i_t \odot g_t && \text{cell state} \\
h_t &= o_t \odot \tanh(c_t) && \text{hidden state}
\end{aligned}
$$

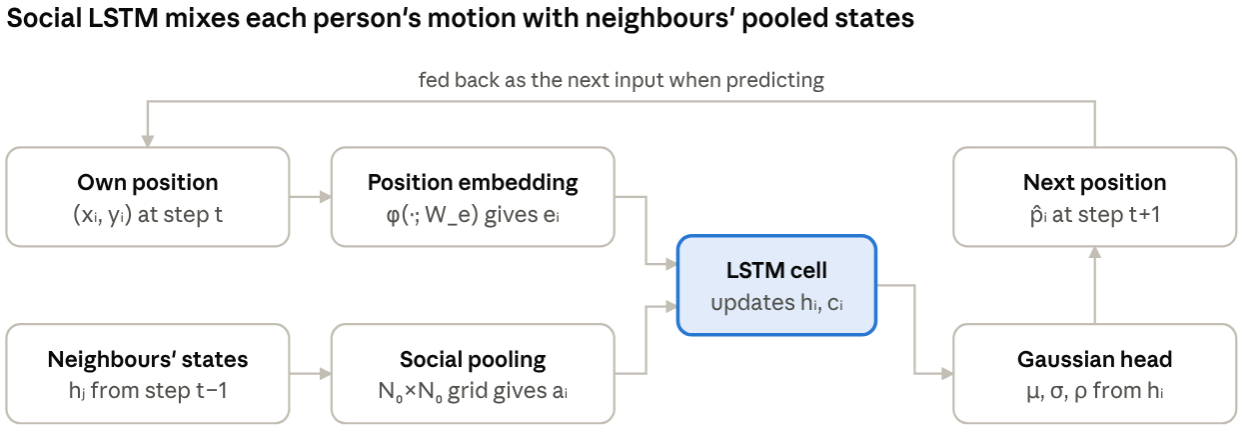
σ is the logistic sigmoid and $\odot$ element-wise multiplication. **[Interpretation]** For a pedestrian, c_t can carry slow quantities such as heading and preferred speed, while h_t is the per-step summary shared with neighbours in Social LSTM.

6.4 Social pooling and spatial neighbourhoods

[Verified] Social LSTM [6] embeds each position, builds a social tensor from neighbours' previous hidden states on an $N_o \times N_o$ grid centred on pedestrian i, embeds it, and feeds both to the LSTM:

$$
\begin{aligned}
e_t^i &= \phi(x_t^i, y_t^i; W_e) \\
H_t^i(m,n,:) &= \sum_{j \in \mathcal{N}_i} \mathbb{1}_{mn}\left[x_t^j - x_t^i,\; y_t^j - y_t^i\right] h_{t-1}^j \\
a_t^i &= \phi(H_t^i; W_a) \\
h_t^i &= \mathrm{LSTM}\left(h_{t-1}^i, e_t^i, a_t^i; W_l\right)
\end{aligned}
$$

$\mathbb{1}_{mn}$ is 1 when neighbour j falls in grid cell (m, n), and φ is a ReLU-activated linear embedding. **O-LSTM** replaces the hidden states by a count, giving an occupancy map $O_t^i(m, n) = \Sigma_j \mathbb{1}_{mn}[\ldots]$.



Social LSTM, one time step for pedestrian i

The only path by which neighbours affect pedestrian i is the pooled tensor a_i; removing it gives Vanilla LSTM, and replacing hidden states with counts gives O-LSTM.

[Interpretation] Other neighbourhood operators in Section 3, in the same notation:

- **Global pooling** (Social GAN): an MLP on each relative position and hidden state, then element-wise max over all j.
- **Attention** (Social Attention, STGAT): $a_t^i = \Sigma_j \alpha_{ij} h_j$, with weights α_{ij} from a softmax over learned scores.
- **Graph convolution** (Social-STGCNN): a weighted adjacency A_{ij} from a distance kernel, then normalised graph convolution.
- **Force-based** (social force, NSP): $a_t^i = \Sigma_j f(p_i - p_j, v_i, v_j)$, a physically shaped function.

6.5 Gaussian output and negative log-likelihood

[Verified] Social LSTM predicts the next position as a bivariate Gaussian with mean $\mu = (\mu_x, \mu_y)$, standard deviations $\sigma = (\sigma_x, \sigma_y)$ and correlation ρ, from a linear layer on h_t, and trains by minimising the negative log-likelihood [6]:

$$\mathcal{N}(\mathbf{p} \mid \boldsymbol{\mu}, \boldsymbol{\sigma}, \rho) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\!\left(-\frac{Z}{2(1-\rho^2)}\right),\quad Z = \frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}$$

$$L^i = -\sum_{t=T_{obs}+1}^{T_{pred}} \log \mathcal{N}\left(\mathbf{p}_t^i \mid \boldsymbol{\mu}_t^i, \boldsymbol{\sigma}_t^i, \rho_t^i\right)$$

[Verified] Graves's mixture-density sequence model keeps scale parameters positive with the exponential function and correlations in $(-1, 1)$ with tanh [33]; Social LSTM's output follows the same pattern. **[Interpretation]** Numerical care, such as clamping ρ away from ± 1, is a known implementation risk.

6.6 Other trajectory losses

$$\begin{aligned}
L_{L2} &= \frac{1}{T_p}\sum_{t} \lVert \hat{\mathbf{p}}_t - \mathbf{p}_t \rVert_2^2 \\
L_{variety} &= \min_{k \in \{1..K\}} \lVert \mathbf{Y} - \hat{\mathbf{Y}}^{(k)} \rVert_2 \\
L_{CVAE} &= -\mathbb{E}_{q(z \mid x,y)}[\log p(y \mid x,z)] + \beta\, \mathrm{KL}\big(q(z \mid x,y) \,\Vert\, p(z \mid x)\big)
\end{aligned}$$

[Verified] Social GAN's variety loss rewards only the best of K samples, which encourages diverse outputs [7]. Trajectron++ and AgentFormer are CVAE-based [13], [19].

6.7 Evaluation metrics: ADE, FDE and beyond

$$\mathrm{ADE} = \frac{1}{N\,T_p}\sum_{i=1}^{N}\sum_{t=T_{obs}+1}^{T_{pred}} \lVert \hat{\mathbf{p}}_t^i - \mathbf{p}_t^i \rVert_2, \qquad \mathrm{FDE} = \frac{1}{N}\sum_{i=1}^{N} \lVert \hat{\mathbf{p}}_{T_{pred}}^i - \mathbf{p}_{T_{pred}}^i \rVert_2$$

- **minADE$_k$ / minFDE$_k$:** the same, using the closest of K samples (K = 20 is common) [7].
- **Collision rate:** share of predictions in which two pedestrians come closer than a benchmark-defined distance [14].
- **[Verified] Definitions are not always consistent.** The original Social LSTM paper described its average displacement error as a mean squared error [6], and the STGAT authors later corrected their paper's ADE description to L2 distance [11]. **[Recommendation]** Our evaluation module will implement the formulas above exactly and state units (metres) in every table.

6.8 Optimisation and backpropagation

Recurrent models are trained by backpropagation through time (BPTT): the network is unrolled over the sequence and gradients of the loss flow back through every step [42]. Parameters are updated with a stochastic optimiser, typically Adam [41]:

$$\theta_{k+1} = \theta_k - \eta \, \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon}$$

m̂ and v̂ are bias-corrected running means of the gradient and squared gradient. **[Interpretation]** Gradient clipping, teacher forcing during training versus autoregressive rollout at test time, and early stopping on a validation scene are the main practical controls.

7. Research gaps

The largest gaps are not missing architectures but missing evidence: dense-crowd data, consistent evaluation and reproducible baselines.

1. **Density gap [Verified].** Most public trajectory datasets are low-density, free-flow scenes; deep models do well there while physics models remain stronger at high density [37]. Truly high-density datasets are still missing [35]. Railway platforms at peak are exactly the under-covered regime.

2. **Reproducibility gap [Verified].** Social LSTM's benefit over Vanilla LSTM is contradicted between papers by the same group [16]; AgentFormer's published numbers changed after a normalisation bug fix [19]; STGAT's paper misdescribed its ADE [11]. Surveys of 200+ methods also call for standardised metrics [31].
3. **Baseline gap [Verified].** A constant velocity model can beat neural models [15], and a Transformer with no interaction terms topped TrajNet [17]. Many papers do not report such baselines. **[Interpretation]** The real contribution of interaction modules is therefore uncertain on standard benchmarks.
4. **Evaluation gap [Verified].** Best-of-N metrics reward a model for one good sample among many, which is unclear for real-time use [16]. Distance metrics ignore physical plausibility; collision metrics were introduced to address this [14].
5. **Uncertainty gap [Verified].** Human motion is multimodal [7]; motion uncertainty is listed as a core open challenge [30]. Deterministic models give one answer with no confidence; generative models give many but are harder to evaluate.
6. **Generalisation gap [Verified].** Neural predictors implicitly learn scene-specific priors that hurt transfer to new scenes [15]. Physics-informed models have shown robustness to 2–5× training density [22], but this has few independent reproductions.
7. **Dense-scene engineering gap [Verified].** On a new dense CCTV benchmark, training time rose by up to 8× and pipelines were sensitive to tracking noise [38].
8. **Context gap [Verified].** Indian railway-station studies we found measure flow and capacity from video (e.g. Palakkad and Trivandrum Central) rather than releasing trajectory datasets [44]. **[Interpretation]** No public Indian station trajectory dataset was found in this review; this is an open question for the supervisor.
9. **Interpretability gap [Verified].** Interpretability is one of five core challenges in recent surveys [30]; black-box predictors are hard to trust for safety decisions. Attention architectures such as Transformers [34] expose weights, but weights alone are not an explanation **[Interpretation]**.

8. Research opportunities

The realistic contribution of this project is careful, reproducible evidence about when interaction modelling helps, applied to station-like crowds — not a new state-of-the-art architecture. All items here are **[Recommendation]**; none is claimed as novel until checked against the literature in Phase 2.

1. **Controlled comparison of interaction mechanisms.** Same data, splits, coordinates, training budget and seeds for kinematic, physics, recurrent, pooled, attention/graph and one hybrid model. Report deterministic and best-of-K results separately. Addresses gaps 2–4.

2. **Density-stratified evaluation.** Report ADE, FDE and collision rate per local-density band, with bands tied to the fundamental diagram [4]. Addresses gap 1; follows the density-classification idea shown by Korbmacher and Tordeux [37].
3. **Interaction-category analysis.** Break errors down by TrajNet++ categories (e.g. leader-following, collision avoidance, groups) [14] to show where pooling or attention actually changes predictions.
4. **Physics-informed reference.** Fit a social force model to the same data and, as an extension, a hybrid that learns force parameters or a residual on top of physics, following the NSP direction [22]. Addresses gaps 1 and 6 and suits a mechanical-engineering report.
5. **Station transfer test.** Train on ETH/UCY, test zero-shot and fine-tuned on Grand Central and, if usable, the Czech terminals [39]. Directly answers RQ4.
6. **Behavioural analysis layer.** Measure speed–density relations, flows, group statistics and lane formation in each dataset, so prediction errors can be explained by crowd state rather than reported as single averages.
7. **Open, documented tooling.** A reproducible pipeline and dashboard that log every run, so all reported numbers can be regenerated — the direct remedy for the reproducibility gap.

9. Preliminary research methodology and next phase

The research supports a staged comparison: strong simple baselines first, then increasingly expressive interaction models, all under one audited protocol, evaluated by accuracy, plausibility and density. Everything in this section is **[Recommendation]** pending supervisor approval.

9.1 Method

1. **Data protocol.** ETH/UCY with leave-one-scene-out; 0.4 s steps; 8 observed, 12 predicted; world coordinates in metres; inputs as relative displacements. Station-like test sets: Grand Central (after homography), Czech terminals (after format audit).
2. **Behaviour analysis.** Per dataset and scene: density, speed, flow, speed–density curve, group and interaction-category statistics.
3. **Models.** The professor's baseline is the reference model; constant velocity and linear predictors are added as sanity baselines; further models only if approved after inspection.
4. **Training controls.** Same embedding and hidden sizes where architectures allow, same optimiser and epoch budget, early stopping on a held-out scene, 3 or more seeds, all configurations logged.

5. **Reproduction gate.** Before new experiments, each re-implemented model is compared with its published numbers under the same protocol; differences are recorded, not hidden.
6. **Evaluation.** ADE, FDE (deterministic) and minADE/minFDE at K = 20 (labelled), collision rate, per density band and per interaction category, mean ± standard deviation over seeds.
7. **Analysis.** Answer RQ1–RQ4 with tables and plotted trajectories; collect failure cases.

9.2 Unresolved questions for the supervisor

- [ ] Is raw video in scope (detection and tracking), or trajectories only?
- [ ] Which reference materials and rubric has the supervisor provided?
- [ ] Is any railway-station footage or trajectory data, especially Indian, available to us?
- [ ] What GPU access exists (local, Colab, institute cluster)?
- [ ] Final deadline, interim reviews and team roles.
- [ ] Grand Central licence terms and homography source; Czech dataset format.
- [ ] Author list of the CrowdTraj preprint [38] to be confirmed before citing it in the final report.

9.3 Recommended next phase

Superseded (11 Nov 2026). The next phase is now baseline inspection and reproduction; see the Research strategy tab.

10. References (IEEE)

All entries were checked against a publisher page, arXiv record, official repository or bibliographic index during this review. Page numbers are given only where a record confirmed them.

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